

E-Health for All: A User-Centered Telemedicine System

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1 Introduction

1.1 Background of the Study

Healthcare access remains one of the most persistent global inequalities, with limited specialist availability and inadequate infrastructure keeping millions from critical care. The World Health Organization estimates that half of the global population lacks access to essential health services, with rural communities disproportionately affected [1], leading to delayed diagnoses and preventable deaths. Telemedicine presents a promising solution by delivering healthcare across distances, connecting remote patients with specialists without requiring travel and making scarce resources available to underserved populations.

In the Philippines, these challenges are especially severe. The country’s geography and urban concentration of medical specialists leave rural populations with limited healthcare options. In response, the Philippine government and health institutions have begun embracing telemedicine to broaden coverage. The National Telehealth Center has launched SMS and web-based platforms connecting rural patients with specialist consultations [2], [3], and various health institutions have adopted electronic medical records to improve continuity of care across facilities [4], [5].

Despite these efforts, significant barriers persist. Poor transportation infrastructure hinders facility access, particularly during emergencies and follow-up care. Unreliable internet connectivity, limited computing resources, and low digital literacy among both patients and healthcare workers further compound these challenges [6], [7]. Existing telemedicine initiatives, while promising, remain fragmented and rarely address the unique technological constraints and cultural context of rural Filipino communities [8], [9].

This research seeks to bridge these gaps by developing an integrated telemedicine platform specifically designed for the selected rural barangays. The platform combines three components, a comprehensive doctors’ database, a patient records management system, and an appointment scheduling system into one user-friendly interface accessible to users with different levels of digital literacy.

This initiative aligns with multiple UN Sustainable Development Goals: good health and well-being (sdg3), by improving healthcare access and quality. It supports industry, innovation, and infrastructure (sdg9), by demonstrating how technology can address healthcare infrastructure gaps in underserved areas. It advances reduced inequalities (sdg10), by examining how digital health solutions reduce disparities in healthcare access across different populations and socioeconomic groups.

1.2 Review of Related Literature

Telemedicine Adoption and Implementation

Telemedicine, the delivery of healthcare services using information and communication technologies, has emerged as a vital solution for healthcare access issues in isolated and underserved communities. Telemedicine allows healthcare providers to assess, diagnose, and treat patients in remote areas by utilizing telecommunications technology to effectively connect medical experts with populations that have limited access to healthcare facilities. It has shown great promise in overcoming healthcare access issues in rural communities around the world.

Research shows that telemedicine consultations promote better clinical decision-making, reduce unnecessary patient transfers, and increase the chances of local admissions instead of costly transfers to far-off hospitals [10], [11]. These advantages apply to many medical fields, from primary care to specialized areas like neurology and emergency medicine [12].

However, the pandemic highlighted the shortcomings of telemedicine: access services at lower rates [13],[14]. Digital literacy among older adults and low-income individuals [6],[7], equal access concerns [15],[16],[17], implementation costs and maintenance [18],[19], infrastructure issues [20],[21],[22], and data security and patient privacy concerns [23].

On the one hand, communities with integrated systems achieve better health outcomes for managing chronic diseases compared to those that use fragmented record-keeping [24]. Patient satisfaction with telemedicine in the Philippines shows encouraging trends [24],[25]. Even senior citizens, increasingly accepted remote consultations [26]. The University of the Philippines Health Service successfully used tailored electronic medical records systems in various settings during COVID-19, ensuring continuity of patient care despite mobility restrictions [4],[3]. Thorough assessments show considerable progress in adoption over time, although significant gaps between different types of facilities and locations persist [27],[28].

Successfully implementing telemedicine requires careful adjustments to local circumstances. Detailed analyses reveal Philippines-specific barriers like governance issues, unclear licensure requirements, unresolved liability problems, and vague reimbursement mechanisms that hinder healthcare provider involvement [2],[9]. Healthcare technology must consider local cultural practices, language preferences, and health-seeking behaviors to ensure meaningful user acceptance in diverse Filipino communities [8]. Creating user-friendly systems is crucial for sustained adoption rather than leaving behind abandoned pilot projects [29].

Patient Information Management System

Patient information management involves the organized collection, storage, and use of health data to support clinical decision-making and ensure continuity of care. Electronic Medical Records (EMR) systems digitize patient health information,

enabling providers to access complete histories, track treatments, identify medication interactions, and coordinate care across specialties. This is especially critical in rural and underserved areas where access to prior care records is limited.

EMR systems significantly improve care coordination, with integrated communities achieving better chronic disease outcomes compared to those with fragmented records [30], [31]. However, while basic adoption rates are high, comprehensive implementation remains lower in rural facilities despite recognized benefits [16], [17]. Technical, organizational, and workflow barriers continue to hinder smooth information sharing [21], [27], [28], and digital literacy gaps among older adults and low-income individuals require targeted design and training interventions [6].

Data security and patient privacy are foundational to EMR trust, requiring encryption, access controls, and audit trails [23]. Integrated systems reduce administrative burdens and improve care delivery [32], [33], [34], while predictive models using patient data enable targeted interventions and better resource allocation [35], [36], supporting a shift toward proactive, preventive primary care [37].

In the Philippines, EMR adoption shows both progress and challenges. COVID-19 visibly impacted system usage in rural health facilities [38], yet implementation improved clinical decision-making and reduced medication errors compared to paper systems [5]. Sustained adoption is linked to training, technical support, and gradual workflow integration rather than rapid rollouts [39], though research representation across specialties, settings, and facility types remains uneven [40]. The absence of centralized provider databases causes referral delays and disjointed care [41], while comprehensive physician databases with specialization and scheduling details enhance diagnostic workflows [42]. Keeping such databases accurate remains challenging, necessitating automated verification and validation processes [43].

Appointment Scheduling in Healthcare

Appointment scheduling systems are essential to healthcare delivery, coordinating patient needs, provider availability, and facility resources. Well-designed systems reduce waiting times, lower missed appointments through automated reminders, improve provider productivity, and ensure urgent cases receive prompt attention. In rural settings, where patients face long travel distances and transportation barriers, efficient scheduling is especially critical for maintaining access and treatment adherence.

Digital scheduling significantly reduces no-show rates compared to offline booking, with automated reminders proving particularly effective [32]. AI-driven tools further improve efficiency by predicting non-attendance and optimizing resource allocation [33], while open-access scheduling reduces no-shows by offering same-day or next-day appointments [34]. Predictive models incorporating patient demographics and appointment history identify high lead time and prior missed appointments as the strongest predictors of non-attendance [35], [36]. Integrated systems also improve patient flow and reduce waiting area congestion in underserved facilities [44], sup-

porting broader proactive primary care initiatives through timely referrals and fewer missed appointments [37], [43].

In the Philippines, digital appointment management in rural health centers has shortened waiting times and reduced missed appointments through better time slot management and automated reminders [45]. However, rural contexts require flexible approaches, as rigid scheduling is ineffective where patients face unreliable transportation or inflexible work schedules. Systems must therefore incorporate easy rescheduling options and reminder methods that function without internet access [46].

1.3 Synthesis of the State-of-the-Art

The detailed collection of works and studies offered valuable insights into the topic. These studies backed their claims with relevant data and detailed information. This section will provide a brief overview of the reviews of relevant literature.

Telemedicine has emerged as a viable solution to geographic barriers, improving clinical decision-making, reducing unnecessary transfers, and expanding specialist access [12], [10], [11]. However, persistent challenges remain, including infrastructure limitations, digital literacy gaps, data security concerns, and Philippines-specific barriers such as unclear licensure, liability, and reimbursement mechanisms [2], [23], [7], [9]. Despite these, large-scale pilots in the Philippines demonstrate feasibility across diverse communities, with patient satisfaction showing encouraging trends even among senior citizens [26], [24], [25], [3]. Successful adoption requires culturally sensitive, user-centered design tailored to local contexts [8], [29].

Patient information management systems significantly improve care coordination and clinical outcomes, yet comprehensive EMR implementation remains lower in rural facilities [17], [16]. Workflow barriers, digital literacy gaps, and data security concerns continue to hinder full adoption [6], [27], [28], [23]. In the Philippines, gradual integration supported by training and technical assistance yields better sustained adoption [39], while the absence of centralized provider databases causes referral delays and fragmented care [42], [41].

Appointment scheduling systems directly reduce no-show rates, improve patient flow, and support proactive primary care [37], [32], [33]. Predictive modeling and AI-driven tools further optimize resource allocation [35], [36], [33]. In Philippine rural health centers, digital scheduling has measurably shortened waiting times [45], though flexible, offline-capable systems are essential given transportation and connectivity constraints [46].

Across all three areas, the literature consistently points to the need for an integrated, context-sensitive platform that unifies patient records, provider information, and appointment scheduling precisely the gap this research addresses.

1.4 Gap Bridged by the Study

This study addresses important gaps in rural healthcare delivery by creating an integrated telemedicine platform specifically for rural barangays in Naga City. Existing telemedicine systems often look at patient management and appointment scheduling separately. This research offers a more complete solution by combining a doctors' database, a patient records management system, and appointment scheduling into one platform. It addresses the unreliable internet connectivity that is common in rural areas. Healthcare workers can keep entering data and accessing patient information even without internet access. Once the connection is restored, the system syncs automatically. By bringing these features together in one practical system designed for resource-limited settings, this research shows how telemedicine solutions can be adapted to meet the real technological, infrastructure, and cultural challenges faced in rural healthcare.

1.5 Theoretical Framework

This study is anchored on the following theories: the Task Technology Fit (TTF) Theory by Goodhue and Thompson, as cited by Thanthrige et al. [47]; the Diffusion of Innovations (DOI) Theory by Rogers, as cited by Tsogbe et al. [48]; and the Technology Acceptance Model (TAM) by Davis, as cited by Holtz et al. [49].

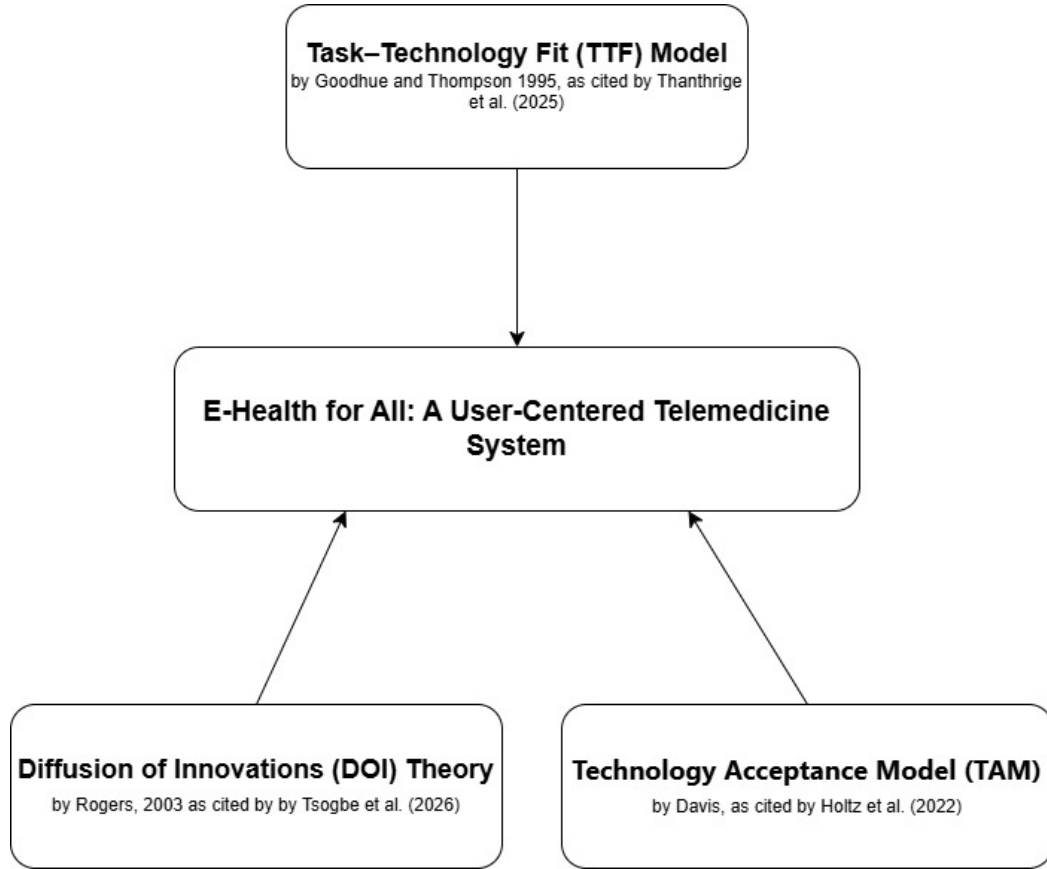


Figure 1.1: Theoretical Framework

The diagram illustrates the theoretical foundations guiding the development of the E-Health for All telemedicine system for improving access in rural communities. Goodhue and Thompson (1995) established that technology is most effectively utilized when its capabilities align with the tasks users must perform. Thanthrige et al. [47] reinforced this principle in a 2025 systematic review, confirming that task-technology alignment remains a critical determinant of healthcare technology adoption. This theory anchors the study by evaluating whether the three core features of E-Health for All digital patient records management, automated health worker scheduling, and online appointment booking fit the actual daily tasks of Panicuason health facility personnel, as a technically sound system that lacks this fit will be underused or abandoned.

Rogers (2003) argued that technology adoption is shaped by five key attributes: relative advantage, compatibility, complexity, trialability, and observability. Tsogbe et al. [48] applied this framework in 2026 within a telehealth adoption context in a developing-region setting, affirming its continued relevance in guiding digital health implementation. This theory guides the study by ensuring that the E-Health for All platform offers a clear operational advantage over the current paper-based system while remaining compatible with the existing workflows of Barangay Panicuason.

Davis (1989) proposed that technology adoption is primarily driven by perceived usefulness the degree to which a system improves work performance and perceived ease of use the degree to which it is free of effort. Holtz et al. [49] applied TAM in 2022 specifically to telemedicine adoption among rural populations, affirming its applicability in this context. This theory directly shapes the acceptability evaluation instrument of the study by measuring whether health facility personnel perceive E-Health for All as useful, easy to use, and trustworthy enough to adopt.

1.6 Conceptual Framework

The Input-Process-Output (IPO) Framework, added with Feedback loop as the fourth component, is the approach utilized in the conceptualization of this study.

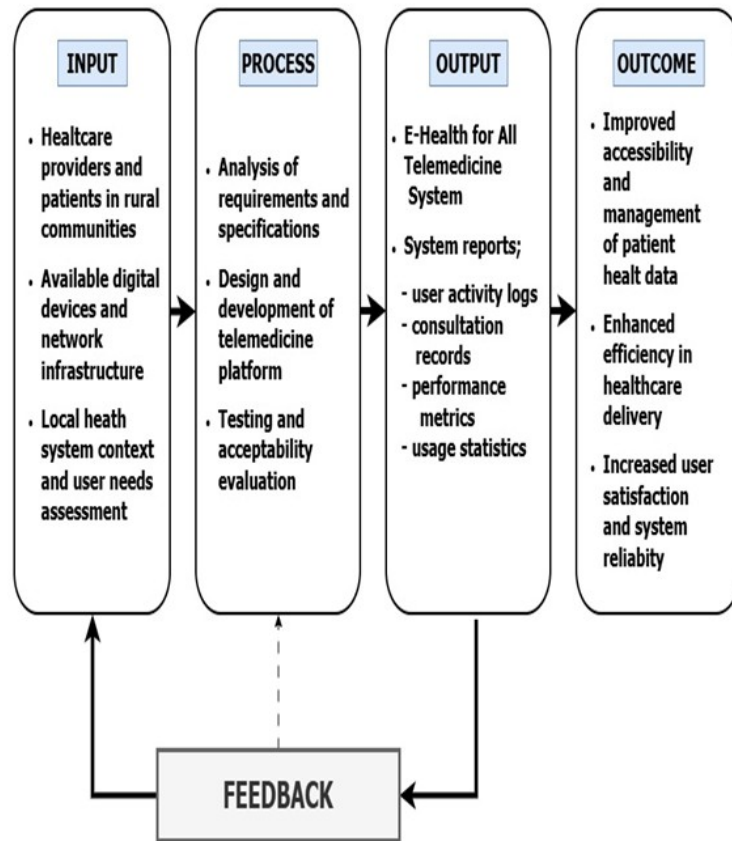


Figure 1.2: Conceptual Framework

The enhanced system paradigm is the approach utilized in the conceptualization of this study. The input came from various sources to gain a complete understanding of rural healthcare needs and readiness for technology, such as document analysis of health records, observations of healthcare-seeking behaviors and travel times, surveys on evaluating user satisfaction and technology skills, and interviews gathering in-depth information about access challenges and provider readiness for

telemedicine services. The processes involved includes, systems analysis and specification for system design, functional testing to ensure adherence to system requirements, usability testing to ensure the system meets the user needs, and acceptability evaluation to validate the system feasibility for deployment. The studys output is a web application with the following features like computer vision, security, authentication, verification, patient check-up history, Qr code security, by using the system it is expected to improve efficiency, achieve fewer documentation and smoother, admin shorter waiting times, through better appointment, enhance user satisfaction, and improve data security of patient information. The continues feedback mechanism will ensure iterative refinement of the system to remain user friendly, operational and align with rural healthcare.

1.7 Statement of the Problem

This study aims to address the problem of healthcare accessibility among rural populations. Specifically, this study seeks to answer the following questions:

1. What are the current practices of rural healthcare providers in rural populations with respect to:
 - (a) Patient records management
 - (b) Doctor scheduling
 - (c) Appointment systems
2. What are the challenges encountered in the current system with respect to:
 - (a) Patient records management
 - (b) Doctor scheduling
 - (c) Appointment systems
3. What system can be designed to address the challenges encountered?
4. What is the level of acceptability of the system in terms of:
 - (a) Usability
 - (b) Reliability
 - (c) Data security

1.8 Assumptions

This research study is anchored on the following key assumptions:

1. Rural healthcare providers in Panicuason, Naga City have specific practices for managing patient records, scheduling doctors, and handling appointments.

2. There are challenges with the current systems for managing patient records, scheduling doctors, and setting up appointments in rural healthcare facilities.
3. A user-centered telemedicine platform can be designed and developed to address these challenges in the current healthcare delivery system.
4. Rural healthcare providers and clinic staff are willing to transition from the current manual practices to the proposed user-friendly telemedicine platform and digital records management system.

1.9 Scope and Delimitation

This study covered the design, development, and initial acceptability evaluation of the E-Health for All telemedicine platform for Barangay Panicuason Rural Health Unit in Naga City, Camarines Sur. The prototype included digital patient health profiles, clinical encounter documentation, vital signs monitoring, medication tracking, appointment booking, automated scheduling, virtual consultation, offline synchronization, prescription scanning, a patient portal, and an integrated health worker database. The 13 health facility personnel 7 Health Workers, 1 Midwife, 1 Encoder, 1 BNS, 1 SFV, 1 CBRS, and 1 BSWA served as respondents through total enumeration. The system was built using PHP, HTML, CSS, and JavaScript with a Supabase cloud database, compatible with Windows, macOS, Linux, Android, and iOS devices.

The study was delimited to Barangay Panicuason only and did not cover other healthcare facilities, long-term performance, cost-effectiveness, laboratory integration, payment and billing, or pharmacy management. External factors such as internet instability, power interruptions, and varying digital literacy levels were outside the scope of this study.

1.10 Significance of the Study

This study benefits the following stakeholders in advancing rural healthcare delivery and digital health innovation:

Patients in Rural Communities. Rural residents gain access to healthcare services without costly travel, reducing time and financial burdens while ensuring continuity of care for chronic conditions.

Rural Healthcare Providers. Healthcare providers benefit from automated records management, streamlined appointment scheduling, and simplified specialist referrals, enabling more efficient and accurate patient care.

Community Health Workers. Frontline healthcare workers gain faster access to patient information, reducing paperwork and allowing greater focus on direct care, remote consultations, and early detection of emerging health trends.

Health Administrators and Local Government Officials. Administrators gain data-driven insights into service utilization, disease prevalence, and resource allocation, along with a replicable model for digital health initiatives in underserved communities.

Technology Developers. Developers gain practical, evidence-based insights into building offline-capable and low-bandwidth healthcare systems designed for users with limited digital literacy.

Academic Researchers. The study provides validated frameworks and tools for designing context-sensitive digital health solutions, addressing existing research gaps on integrated telemedicine platforms in resource-limited environments.

1.11 Definition of Terms

This section provides clear and precise definitions of key terms used throughout the study, both conceptually and operationally.

E-health. E-health refers to the use of information and communication technologies in healthcare delivery, including telemedicine, electronic medical records, and digital health platforms [1]. In this study, it encompasses all electronically delivered health services through the proposed telemedicine platform for rural communities.

E-health for All. E-health for All is a principle that advocates equal access to digital health technologies regardless of location, income, or technical ability [50]. In this study, it refers to the commitment to developing inclusive and user-friendly telemedicine solutions for rural communities in Naga City.

Telemedicine. Telemedicine is the delivery of healthcare services and clinical information through communication technologies to overcome geographical barriers [51]. In this study, it refers to the proposed remote healthcare delivery system developed for Naga City.

User-Centered Design. User-centered design is a development methodology that focuses on the needs, preferences, and limitations of end-users throughout the design process [52]. In this study, it refers to the participatory design approach used in developing the telemedicine platform.

Rural Healthcare. Rural healthcare refers to the provision of medical services in isolated or underserved areas with limited infrastructure and specialist access [53]. In this study, it refers to healthcare delivery in rural barangays of Naga City.

Telehealth Platform. A telehealth platform is a digital healthcare ecosystem integrating provider directories, patient information systems, and scheduling tools to facilitate remote healthcare delivery [54]. In this study, it refers to the web-based system developed for Naga City.

Healthcare Accessibility. Healthcare accessibility refers to the ease with which individuals obtain necessary health services considering physical, geographical,

economic, and organizational factors. In this study, it refers to how rural residents of Naga City access healthcare services through the proposed telemedicine platform.

2 Methodology

This section outlines the research methodology. It covers the research design, participants, location, data collection methods and tools, ethical considerations, statistical analysis, project timeline, and validation methods to ensure the reliability of the findings.

2.1 Research Design

This study adopted a developmental-descriptive research design to document current healthcare practices, identify operational challenges, and develop a telemedicine solution suited to the needs of Barangay Panicuason RHU. Data were collected through surveys, interviews, observations, and document analysis, following four phases: preparation, assessment, data collection, and analysis.

2.2 Population and Sampling

The respondents were the 13 health facility personnel of Barangay Panicuason Rural Health Unit, selected through total enumeration as primary end-users of the proposed system. The respondents comprised 7 Dental Health Workers, 1 Midwife, 1 Encoder, 1 Barangay Nutrition Scholar (BNS), 1 Supplemental Feeding Volunteer (SFV), 1 Community-Based Rehabilitation Specialist (CBRS), and 1 Barangay Social Welfare Aide (BSWA).

For the succeeding phase involving rural community residents as secondary respondents, purposive sampling will be applied, limited to residents with direct experience accessing the barangay health facility. The required sample size will be determined using Slovin's formula:

$$n = \frac{N}{1 + Ne^2} \quad (2.1)$$

where:

- n = required sample size
- N = total population of Barangay Panicuason
- e = margin of error (set at 0.05 or 5%)

This formula helps ensure our sample size is adequate to represent the barangay population while maintaining statistical reliability.

2.2.1 Research Locale

The study was conducted at Barangay Panicuason Rural Health Unit in Naga City, Camarines Sur, a barangay situated outside the urban center and characterized

by limited specialist availability, long travel distances to medical facilities, and insufficient healthcare infrastructure. The facility served as the primary healthcare provider for residents of Barangay Panicuason, offering basic services including primary care, maternal and child health, immunization, and disease prevention, and referring cases requiring specialized care to hospitals in Naga City proper.

2.2.2 Data Gathering Procedure

This section describes the methods and tools the researchers used to capture data from stakeholders essential to this study.

Preparation Phase. The researchers secured formal permissions from barangay officials and local health officers, prepared consent forms, and coordinated with community leaders to schedule observation and interview dates within the first two weeks.

Document Analysis. The researchers examined health records, referral systems, and community health profiles from barangay health stations to document current healthcare access patterns and identify existing challenges.

Direct Observation. The researchers observed and recorded how the health facility personnel managed patient consultations, handled records, and coordinated scheduling and appointments in their actual work setting.

Surveys. A structured questionnaire was administered to all 13 health facility personnel to gather data on current practices, encountered challenges, importance of proposed system features, and level of system acceptability.

Interviews. One-on-one interviews lasting 30 to 45 minutes were conducted with the 13 health facility personnel of Barangay Panicuason Rural Health Unit to explore current healthcare practices, operational challenges, and readiness for a telemedicine solution.

Data Analysis and Documentation. Survey responses were tallied and computed using weighted mean and frequency distribution, while interview data were organized by themes and used as the basis for the final findings and recommendations.

2.3 Research Instrument

The study was used various research instruments to collect accurate and relevant data from participants. These instruments are designed to gather both qualitative and quantitative information to gather specific information that will help us understand the current healthcare access situation and identify what is needed for an effective telemedicine platform.

Interview Guides. Researchers will prepare interview guides for rural residents of Barangay Panicuason covering healthcare experiences, travel distance to facilities, medical costs including transportation, common health problems, mobile and internet access, technology comfort level, and desired telemedicine features. For health center staff, guides will cover record-keeping systems, data management chal-

allenges, internet connectivity, available devices, and infrastructure needs. All guides will include follow-up questions to explore key points further.

Survey Questionnaires. For healthcare providers and health workers, the questionnaire covers demographic profile, years of experience, and current practices in patient records management, doctor scheduling, and appointment systems using a 5-point frequency scale. It also addresses challenges encountered in each of these areas, the importance of proposed system features such as digital record storage, automated scheduling, online appointment booking, and role-based access control, and the perceived level of system acceptability in terms of usability, reliability, and data security.

2.4 Ethical Considerations

This research will follow strict ethical guidelines to protect all participants and ensure responsible handling of operational data. The following measures will be implemented:

Voluntary Participation. Participation was completely voluntary, with no pressure or undue incentives influencing the decision to join. Participants may skip any question or withdraw at any time without consequences or effect on their access to healthcare services.

Informed Consent. All participants received detailed information covering the study’s purpose, data collection and use, potential risks and benefits, confidentiality measures, and the right to withdraw before agreeing to participate. Those who agreed signed an informed consent form.

Parental/Guardian Consent and Minor Assent. For participants under 18, written parental or guardian consent will be obtained prior to any research involvement, and minors aged 7–17 will provide written assent after receiving age-appropriate study information. Minors may decline participation even if parents have given consent.

Consent from Barangay Health Workers (BHWs). BHWs will sign informed consent forms acknowledging permission to collect and use their personal information, professional insights, and community health observations. The form will clarify how their contributions will be used and ensure their information is protected under data privacy regulations.

Privacy and Confidentiality. Individual identities were not revealed without explicit written permission, and all interview recordings and transcripts were stored in password-protected files accessible only to the research team. Researchers will not photograph or record any actual patient medical records containing private health information.

Data Protection. All research data will be handled in accordance with the Data Privacy Act of 2012 (Republic Act No. 10173), with personal identifiers removed

and survey responses kept anonymous. All data will be securely destroyed after thesis completion and any required retention period.

Minimizing Risk. Research procedures present minimal risk, and researchers will travel to participants' communities to reduce inconvenience. If participants feel uncomfortable during interviews, sessions will be paused or ended immediately with appropriate referrals provided if needed.

Cultural Sensitivity. Researchers will approach all interactions with cultural humility and respect for local practices, collaborating with community leaders to ensure research activities are culturally appropriate and scheduled at convenient times.

Researcher Responsibilities. The research team is committed to accurately recording and reporting all data without fabrication or selective omission, properly citing all sources, and honestly reporting the research process including challenges and limitations. Findings will be shared with participating communities upon request.

2.4.1 Statistical Tools

This section explains how we will analyze and interpret the collected data to answer the research questions.

Descriptive Statistics

The study employed descriptive statistical methods to summarize and describe the characteristics of the sample and their responses. The following descriptive statistics were used: percentage, frequency distribution, and weighted mean.

Percentage. To show the proportion of respondents who chose each answer option, we will calculate percentages using the formula:

$$\text{Percentage} = \frac{f}{N} \times 100$$

where:

f = frequency or number of responses for a specific option

N = total number of respondents

For example, if 45 out of 60 rural residents say they want mobile access to health records, the percentage would be calculated as $(45/60) \times 100 = 75\%$.

Frequency Distribution. We will create frequency tables displaying how many respondents selected each response option. For instance, for a satisfaction question with a 5-point scale, the frequency distribution might be presented as:

Weighted Mean. For Likert-scale items where we want to account for the frequency of each response, we will calculate the weighted mean using the formula:

Table 2.1: Frequency Distribution

Response Category	Number of Respondents	Percentage
Very Satisfied	5	8.3%
Satisfied	25	41.7%
Neutral	15	25.0%
Dissatisfied	10	16.7%
Very Dissatisfied	5	8.3%
Total	60	100%

$$\text{Weighted Mean} = \frac{\Sigma(W \times f)}{N}$$

where:

W = weight or scale value (1–5 for a 5-point Likert scale)

f = frequency or number of respondents who chose that value

N = total number of respondents

Σ = summation symbol

2.5 Statistical Treatment of Data

The data gathered will be interpreted using a five-point Likert scale to measure the respondents' attitudes, opinions, perceptions, and preferences regarding the user-centered telemedicine system.

Interpretation Scale. To interpret the weighted mean values obtained from Likert-scale questions, the following scale will be used

Table 2.2: Frequency Distribution

Response Category	Number of Respondents	Percentage
Very Satisfied	5	8.3%
Satisfied	25	41.7%
Neutral	15	25.0%
Dissatisfied	10	16.7%
Very Dissatisfied	5	8.3%
Total	60	100%

2.6 System Design and Architecture

The system design process employed several modeling techniques such as use case and system workflows to map user interactions and the system architecture.

These techniques directly captured the functional requirements, user actions, and step-by-step procedures within the system. This ensured a thorough understanding for designing a robust and efficient telemedicine platform.

Use Case

The use case diagram in the study showed the specific interactions between patients, healthcare providers, and system administrators with the system. It defined the actions between these actors and the use cases, along with the functional requirements. The use case model for this study was shown in Figure 2.1.

In the use case diagram, there were three actors: patients, doctors, and system administrators. All actors can register and log in to the system. Patients and doctors utilize a mobile application, while system administrators access a dedicated website application.



Figure 2.1: Use Case Diagram

System Workflows

The system workflows illustrate the sequential processes and decision points for key operations within the E-Health for All Telemedicine System. These workflows ensure standardized procedures and guide users through various system functionalities.

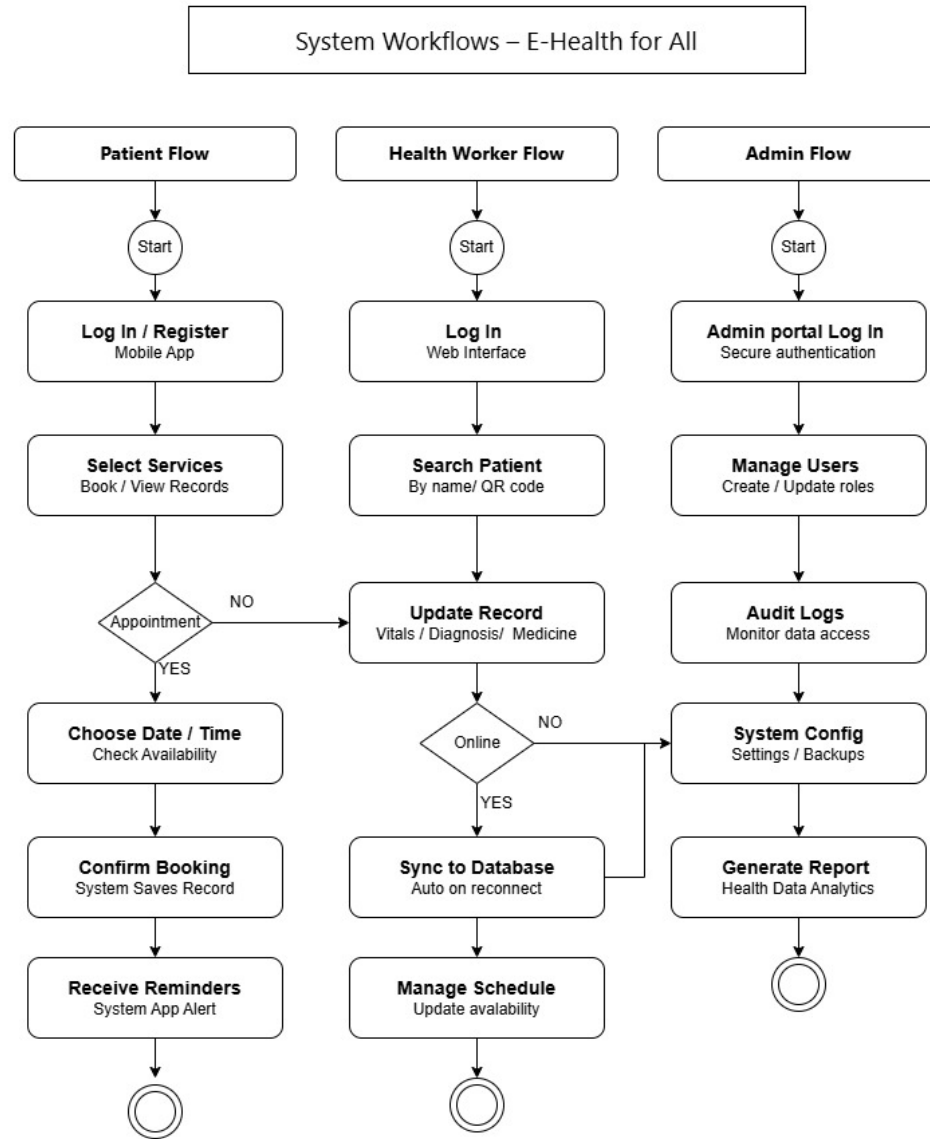


Figure 2.2: System Workflow Diagram

Patient Flow. The patient workflow begins with login or registration. After authentication, patients can select services, check appointment availability, choose dates and times, confirm bookings, receive reminders, and complete their system appointments. This streamlined process ensures efficient access to healthcare services.

Health Worker Flow. Health workers log into the web interface, search patients by name or QR code, update medical records, manage diagnosis and prescriptions, sync data to the database, and complete patient consultations. This workflow maintains accurate and up-to-date patient information.

Admin Flow. Administrators access the admin portal through secure authentication, manage users by creating or deactivating accounts, conduct audit log

reviews, perform system configuration, generate analytical reports on health data, and monitor overall system performance. These workflows ensure proper governance and system integrity.

2.6.1 System Architecture

The E-Health for All Telemedicine System architecture, shown in Figure 5, connects three stakeholder groups healthcare providers, rural residents, and system administrators through a layered structure spanning user devices, connectivity, application services, integration, and data storage. The system is built around three core components in the Application Layer: (1) a web-based platform for healthcare providers, offering secure login, patient records management, clinical documentation, and digital prescription and certificate generation; (2) a mobile application for rural residents, enabling appointment scheduling, medical record access, video consultations, SMS reminders, health education content, and emergency contacts; and (3) a doctor and health worker scheduling system, maintaining provider profiles including specializations, availability, and consultation details

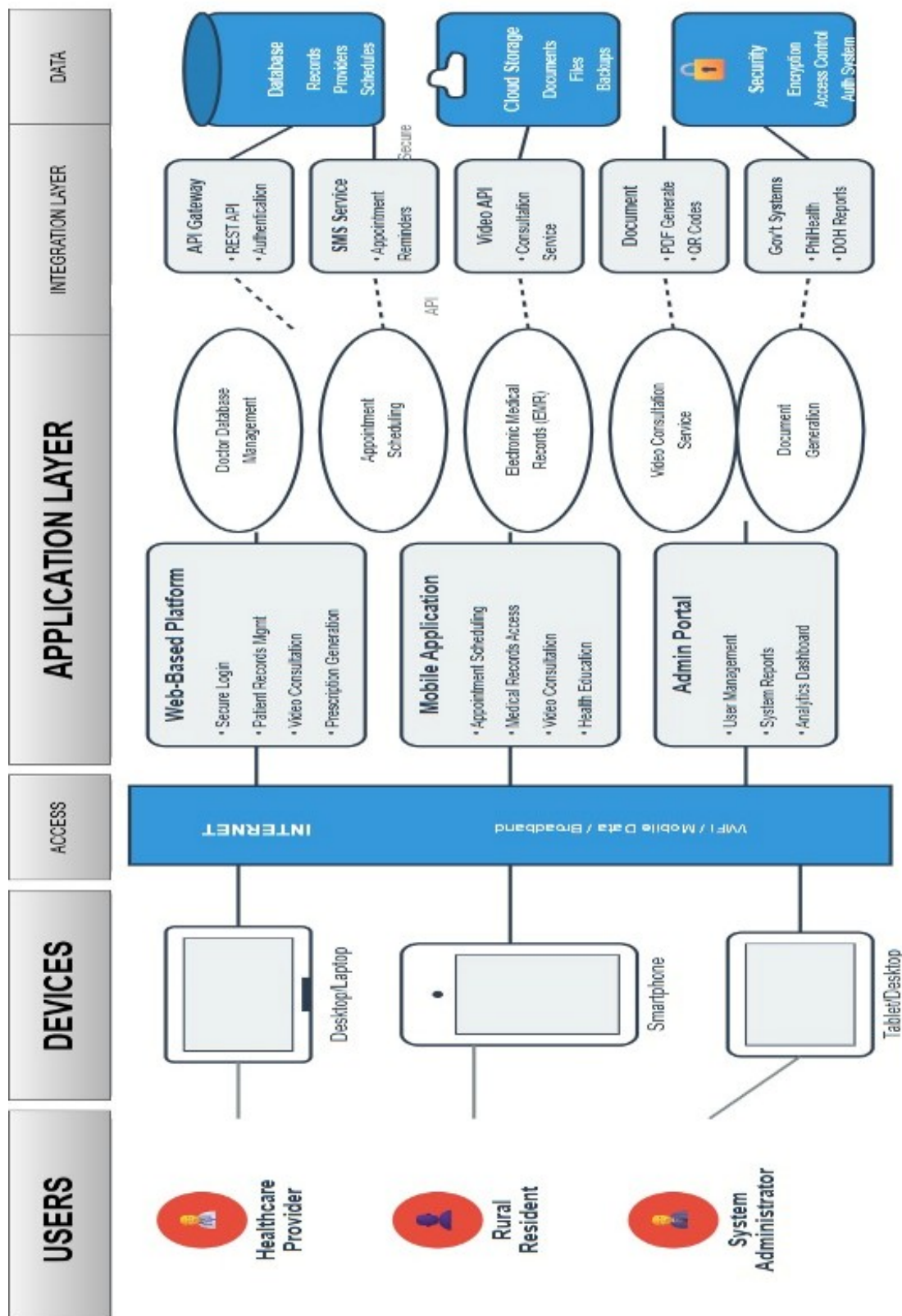


Figure 2.3: System Design Architecture of E-Health for All System

2.6.2 Data Flow Diagrams

DFD Level 0

The Context Diagram shows the entire data flow of the E-Health for All Telemedicine System, where four external entities — Patient, Doctor, BHW, and Admin exchange certain data with the system. This diagram serves as a foundation for designing the system. The data coming into and out of the system is clearly represented.

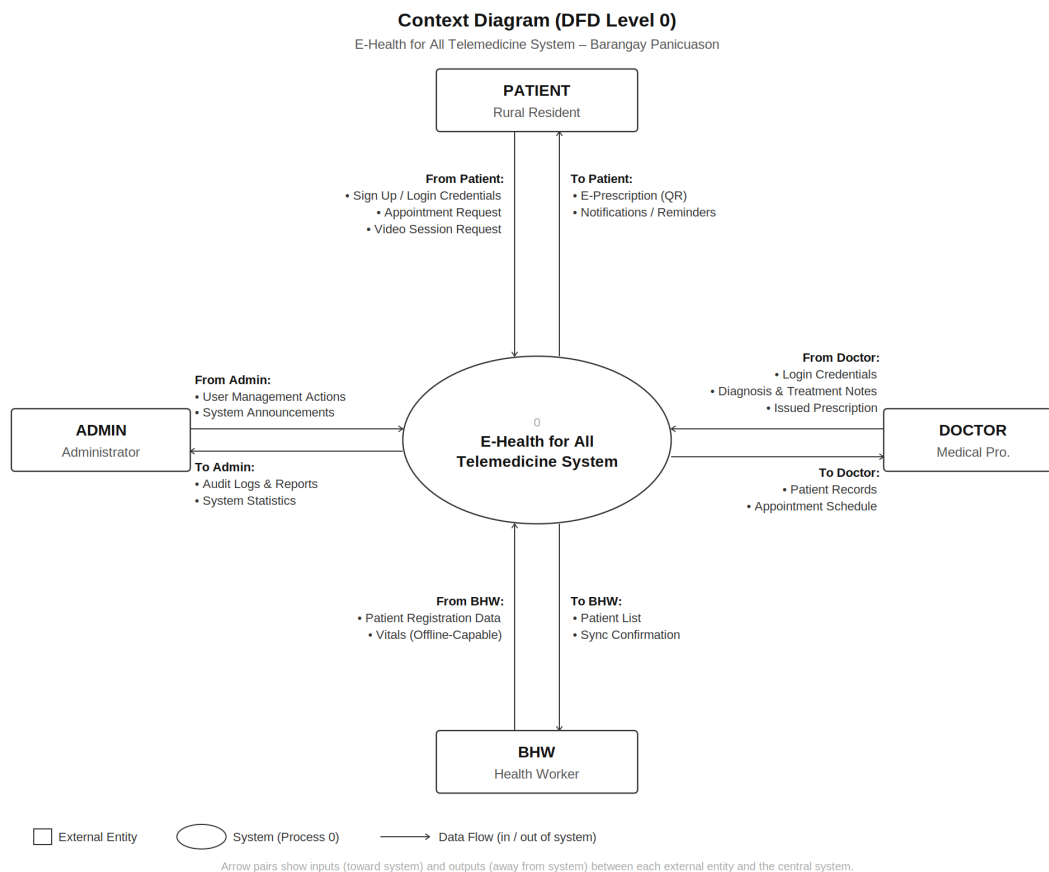


Figure 2.4: DFD Level 0

DFD Level 1

The DFD Level 1 data flow diagram further breaks down the system into its main functions. For now we're comprising eight major processes that Health Connect features on its web and mobile platforms, which included Registration & Authentication, Appointment Management, Video Consultation, Medical Record Management, e-Prescription, Vitals Management, Notification Management, and Audit & Security. The processes outline the movement of information among the various external entities and databases.

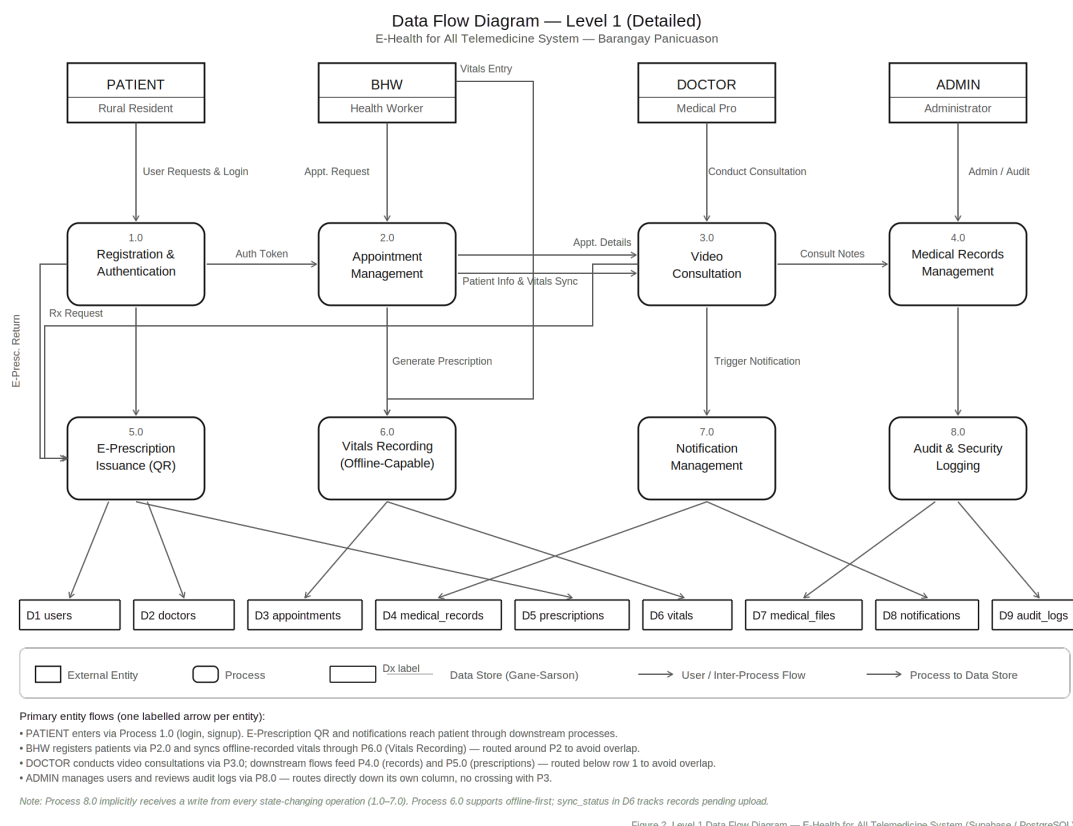


Figure 2.5: DFD Level 1

Entity Relationship Diagram

The ERD visualizes each entity within the database design of the system, which is characterized by nine (9) entities, namely Users, Doctors, Appointments, Medical Records, Prescriptions, Vitals, Notifications, Audit Logs, and Medical Files. It further shows how these entities are linked through their relationships and foreign keys, thus representing the logical arrangement of the database within the system.

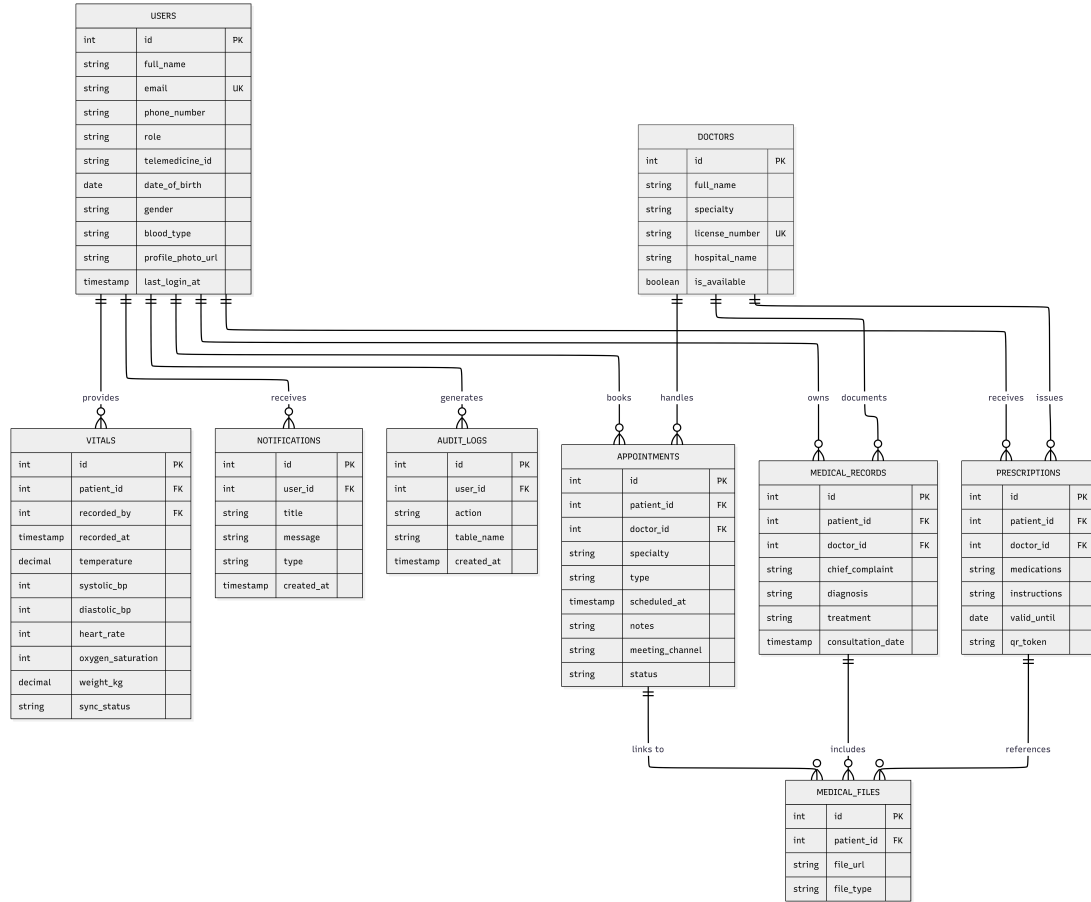


Figure 2.6: Entity Relationship Diagram

2.7 Validation and Testing Methods

To ensure the comprehensive validation and testing of the system's quality and reliability, multiple testing methodologies were employed throughout the development process.

Functionality Testing. Unit testing verified individual components using test cases covering normal inputs, edge cases, and invalid inputs. Integration testing confirmed that data flowed correctly between modules and that integrated components worked together as expected. This approach ensured that each system component functioned correctly both independently and as part of the integrated system.

User Acceptance Testing. This involved healthcare providers evaluating whether the system met their needs through a structured checklist validating that all functional requirements were implemented. Healthcare workers and administrators participated in real-world scenario testing to verify that the system addressed their operational needs and workflow requirements.

Performance Testing. Load testing simulated 10, 20, and 30 concurrent users to ensure acceptable performance under typical and peak loads. Response

time was tracked with a target of under 3 seconds for most operations, ensuring a smooth user experience. Offline functionality testing confirmed data entry and retrieval capabilities without internet connection, which is critical for rural healthcare settings with intermittent connectivity.

Security Testing. Access control verification confirmed users could only access role-appropriate features and that unauthorized access attempts were blocked. Data encryption validation confirmed sensitive information was encrypted in transit via HTTPS and at rest in the database. Passwords were hashed using strong algorithms, and sessions were tested for appropriate expiration and hijacking prevention to protect patient confidentiality and data integrity.

Usability Testing. The think-aloud protocol was conducted with 5 to 8 representative users who verbalized their thoughts while completing tasks as researchers observed and recorded difficulties and errors. Metrics included task completion rates, time on-task, error rates, and post-task satisfaction ratings. This qualitative and quantitative approach provided insights into user experience and identified areas for interface improvement.

3 Results and Discussions

This chapter presents the findings gathered from the survey administered to the health facility personnel of Barangay Panicuason Rural Health Unit in Naga City. Results are organized according to the specific research objectives: (1) current healthcare practices, (2) challenges encountered in the current system, (3) system design plans as the prototype response to identified challenges, and (4) the level of acceptability of the initial prototype in terms of usability, reliability, and data security. Data are presented using weighted mean scores and frequency distributions, discussed in relation to the reviewed literature, and linked to the theoretical frameworks guiding the study.

3.1 Profile of Respondents

A total of 13 health facility personnel completed the survey questionnaire during the data-gathering phase. Table 3.1 presents the demographic profile of the respondents.

Table 3.1: Demographic Profile of Survey Respondents

Category	Frequency (f)	Percentage (%)
A. Sex		
Female	12	92.3%
Male	1	7.7%
Total	13	100%
B. Position / Designation		
Dental Health Worker (DHW)	7	53.8%
Midwife	1	7.7%
Encoder	1	7.7%
Barangay Nutrition Scholar (BNS)	1	7.7%
Community Based Rehabilitation Specialist (CBRS)	1	7.7%
Barangay Social Welfare Aide (BSWA)	1	7.7%
Total	13	100%
C. Age Bracket		
Below 25 years old	1	7.7%
25 – 34 years old	4	30.8%
35 – 44 years old	6	46.2%
45 years old and above	2	15.4%
Total	13	100%
D. Years of Experience		
Less than 1 year	0	0%
1 – 3 years	1	7.7%
4 – 6 years	3	30.77%
7 – 9 years	0	0%
10 years and above	8	61.54%
Total	13	100%

The respondents were predominantly female (92.3%), consistent with the largely female composition of community health workforces in Panicuason. The majority were aged 35–44 (46.2%), indicating a relatively young and potentially technology-receptive group. In terms of work experience, most respondents had 10 years or more of service (61.54%), followed by those with 4–6 years (30.77%), while only a small proportion had 1–3 years (7.69%) and none fell within the 7–9 year range. This suggests that the respondents generally possess substantial professional experience, enabling them to provide informed and reliable assessments of current operational practices. These demographics align with the Task–Technology Fit (TTF) theory’s premise that technology adoption is influenced by users’ experiential familiarity with their work tasks [42].

3.1.1 Current Practices of Rural Healthcare Providers

Research Question 1 asked: ‘What are the current practices of rural healthcare providers with respect to patient records management, doctor and health worker scheduling, and appointment systems?’ Table 3.2 presents the weighted mean scores for each indicator.

Scale: 5 = Always (4.21–5.00), 4 = Often (3.41–4.20), 3 = Sometimes (2.61–3.40), 2 = Rarely (1.81–2.60), 1 = Never (1.00–1.80)

Indicator	WM	Interpretation
A. Patient Records Management		
2.1 Records documented immediately after each consultation	4.70	Always
2.2 Records stored in an organized and accessible manner	4.60	Always
2.3 Records updated regularly to reflect current health status	4.50	Always
2.4 Records retrieved quickly and accurately when needed	4.60	Always
Overall Weighted Mean — Patient Records Management	4.60	Always
B. Doctor and Health Worker Scheduling		
2.5 Schedules planned and communicated in advance	4.20	Often
2.6 Availability clearly recorded and accessible to staff	4.20	Often
2.7 Schedule changes updated and communicated promptly	4.20	Often
2.8 Backup/replacement arranged when scheduled worker is unavailable	4.50	Always
Overall Weighted Mean — Doctor Scheduling	4.28	Often
C. Appointment Systems		
2.9 Appointments arranged in an orderly and systematic manner	4.50	Always
2.10 Patients informed of appointment schedules in advance	4.50	Always
2.11 Appointment records maintained accurately to avoid double-booking	4.30	Often
2.12 Patients attended within a reasonable waiting time	4.20	Often
Overall Weighted Mean — Appointment Systems	4.38	Often

Table 3.2: Current Practices of Rural Healthcare Providers

The results revealed consistently high practice scores across all three domains. Patient records management registered the highest overall weighted mean of 4.60 (*Always*), indicating that health facility personnel demonstrate strong commitment to documenting, organizing, and retrieving patient records. Notably, records are

documented immediately after consultations (WM = 4.70), reflecting adherence to standard clinical documentation protocols.

Doctor and health worker scheduling obtained an overall WM of 4.28 (*Often*), with backup arrangement scoring highest at 4.50 (*Always*), suggesting that contingency planning is well-established at the facility. Appointment systems averaged 4.38 (*Often*), with systematic arrangement and advance patient notification both scoring 4.50 (*Always*). These high practice scores align with findings by Santos, Mendoza, and Cruz [5], who noted that Philippine rural health units have established basic operational protocols, though implementation quality varies significantly between facilities.

These findings support the Task Technology Fit theory [55] in that the existing manual practices are well-established and purposive. A digital system that replicates and enhances these existing workflows rather than replacing them is more likely to achieve strong adoption among health workers who are already familiar with structured documentation and scheduling routines.

3.1.2 Challenges Encountered in the Current System

Research Question 2 asked: “What are the challenges encountered in the current system?” Table 3.3 presents the weighted mean scores for each challenge indicator.

Scale: 5 = Always (4.21–5.00), 4 = Often (3.41–4.20), 3 = Sometimes (2.61–3.40), 2 = Rarely (1.81–2.60), 1 = Never (1.00–1.80)

Challenge Indicator	WM	Interpretation
A. Patient Records Management		
3.1 Patient records are difficult to organize and manage	3.20	Sometimes
3.2 Patient records are missing or incomplete	2.80	Sometimes
3.3 Retrieving a patient’s record takes too much time	3.00	Sometimes
3.4 Records are lost or damaged due to current storage method	2.80	Sometimes
Overall Weighted Mean — Patient Records Challenges	2.95	Sometimes
B. Doctor and Health Worker Scheduling		
3.5 Scheduling conflicts among doctors/health workers occur	3.00	Sometimes
3.6 No clear or updated record of availability	2.90	Sometimes
3.7 Health workers assigned to multiple duties simultaneously	2.70	Sometimes
3.8 Schedule changes not communicated to concerned staff	2.70	Sometimes
Overall Weighted Mean — Scheduling Challenges	2.83	Sometimes
C. Appointment Systems		
3.9 Patients experience long waiting times before being attended to	3.00	Sometimes
3.10 Delays in processing patient appointments	2.90	Sometimes
3.11 Double-bookings or missed appointments occur	2.70	Sometimes
3.12 No organized system for tracking or confirming appointments	3.00	Sometimes
Overall Weighted Mean — Appointment Challenges	2.90	Sometimes

Table 3.3: Challenges in Current Healthcare Practices

All challenge indicators were rated in the “Sometimes” range (WM = 2.61–3.40), confirming that while problems were not constant, they occurred with enough regularity to warrant systematic intervention. The most frequently cited challenge across domains was the difficulty in organizing and managing patient records (WM = 3.20), consistent with findings by D’Amore et al. [16] that paper-based record systems in rural health facilities are particularly susceptible to organizational lapses due to high patient volumes and limited administrative staffing.

Scheduling challenges averaged 2.83, with scheduling conflicts (WM = 3.00) and unclear availability records (WM = 2.90) as primary concerns. This aligns with Mendoza, Cruz, and Santos [39], who found that the absence of a centralized scheduling system in Philippine rural health units contributes to coordination failures, particularly during peak consultation periods. Appointment system challenges averaged 2.90, with the absence of a tracking system (WM = 3.00) and long patient waiting

times (WM = 3.00) as the most persistent issues. Del Rosario [45] similarly found that the lack of digital appointment management in Philippine rural health centers contributes significantly to extended waiting times and patient dissatisfaction.

These “Sometimes” ratings, while not alarming individually, collectively defined a pattern of recurring inefficiency that compounds over time. They validated the need for a digital solution that addresses record organization, scheduling conflict detection, appointment tracking, and automated reminders — all of which are core features of the proposed E-Health for All system.

3.1.3 System Design and Plans for the Prototype

Research Question 3 asked: “What system can be designed to address the identified challenges?” This section presents the importance ratings for proposed system features. Table 3.4 describes the design plans for the prototype, including the use case, workflow, and database schema.

Scale: 5 = Very Important (4.21–5.00), 4 = Important (3.41–4.20), 3 = Moderately Important (2.61–3.40), 2 = Slightly Important (1.81–2.60), 1 = Not Important (1.00–1.80)

Proposed System Feature	WM	Interpretation
4.1 Digital storage and easy retrieval of patient records	4.50	Very Important
4.2 Automated doctor/worker scheduling with conflict detection	4.50	Very Important
4.3 Online patient appointment booking	4.60	Very Important
4.4 Automated appointment reminders and notifications	4.80	Very Important
4.5 Secure login and role-based access control	4.90	Very Important
Overall Weighted Mean	4.66	Very Important

Table 3.4: Importance of Proposed Features

All five proposed features received “Very Important” ratings, with an overall weighted mean of 4.66. Secure login and role-based access control (WM = 4.90) ranked highest, reflecting a strong awareness among health workers of data privacy obligations under the Data Privacy Act of 2012. Automated reminders (WM = 4.80) ranked second, directly addressing the challenge of missed appointments (Item 3.11, WM = 2.70) and long waiting times (Item 3.9, WM = 3.00). Online appointment booking (WM = 4.60) and digital storage (WM = 4.50) also scored highly, validating the core functional design of the proposed system. These ratings are consistent with the TAM framework [56], where perceived usefulness — as evidenced by the high importance ratings — is a primary determinant of technology adoption intention.

Bibliography

- [1] World Health Organization. *eHealth*. <https://www.emro.who.int/health-topics/ehealth/>. 2023.
- [2] A. D. Macariola et al. “Breaking barriers amid the pandemic: The status of telehealth in Southeast Asia and its potential as a mode of healthcare delivery in the Philippines”. In: *Frontiers in Pharmacology* 12 (2021), p. 754011. DOI: 10.3389/fphar.2021.754011.
- [3] A. Pickering et al. “A building blocks approach to implementing a telehealth clinic model to improve primary care access in the Philippines: A large-scale pilot project”. In: *Telehealth Medicine Today* 9.1 (2024). DOI: 10.30953/tmt.v9.596.
- [4] Philippine Primary Care Studies. “Electronic medical records implementation in Philippine health facilities during the COVID-19 pandemic: A multi-site evaluation”. In: *Philippine Journal of Primary Care* 18.1 (2024), pp. 12–28.
- [5] R. P. Santos, F. A. Mendoza, and M. L. Cruz. “Digital transformation of rural health units: Electronic medical records implementation and impact”. In: *Philippine Journal of Health Informatics* 12.2 (2021), pp. 67–83.
- [6] A. Abdul et al. “A review of challenges and opportunities in implementing health informatics in rural healthcare settings”. In: *International Medical Sciences Research Journal* 4.5 (2024), pp. 606–631.
- [7] M. A. Castillo. “Digital literacy barriers in rural Philippine healthcare: Implications for telemedicine adoption”. In: *Asian Journal of Health Information Management* 17.3 (2023), pp. 112–129. DOI: 10.1080/ajhim.2023.567890.
- [8] J. P. Cruz. “Cultural adaptation of digital health technologies in rural Philippines: A framework for user-centered design”. In: *International Journal of Medical Informatics* 168 (2022), p. 104583. DOI: 10.1016/j.ijmedinf.2022.104583.
- [9] R. S. Domingo, A. B. Santos, and L. M. Reyes. “Offline capabilities in telemedicine platforms: Addressing connectivity challenges in rural Philippines”. In: *Journal of Telemedicine and Telecare* 29.8 (2023), pp. 612–623. DOI: 10.1177/1357633X211045678.
- [10] N. Natafghi et al. “Telehealth in critical access hospitals: A mixed-methods study of emergency care telemedicine”. In: *Journal of Rural Health* 37.3 (2021), pp. 478–488. DOI: 10.1111/jrh.12370.
- [11] T. McCormick. “Teleneurology implementation in rural critical access hospital”. In: *Rural Health Information Hub Case Study* (2021).
- [12] V. C. Ezeamii et al. “Revolutionizing healthcare: How telemedicine is improving patient outcomes and expanding access to care”. In: *Cureus* 16.7 (2024), e63881. DOI: 10.7759/cureus.63881.

- [13] C. Chu et al. "Rural telemedicine use before and during the COVID-19 pandemic: Repeated cross-sectional study". In: *Journal of Medical Internet Research* 23.4 (2021), e26960. DOI: 10.2196/26960.
- [14] A. Kichloo et al. "Telemedicine, the current COVID-19 pandemic and the future: A narrative review". In: *Family Medicine and Community Health* 8.3 (2020), e000530. DOI: 10.1136/fmch-2020-000530.
- [15] R. M. Fairchild. "Telehealth adoption barriers among rural healthcare facilities during COVID-19 pandemic". In: *Rural Health Quarterly* 14.3 (2019), pp. 127–145.
- [16] J. D. D'Amore et al. "Electronic health records in rural hospitals: Challenges and opportunities". In: *Journal of Rural Health* 37.2 (2021), pp. 345–359.
- [17] A. J. Anzalone, R. L. Begay, and K. A. Williams. "Lower electronic health record adoption and interoperability in rural versus urban physician participants: A cross-sectional analysis". In: *BMC Health Services Research* 25.1 (2025), p. 123.
- [18] K. S. Zachrison, J. V. Richard, and A. Mehrotra. "Paying for telemedicine in emergency care". In: *Annals of Emergency Medicine* 78.2 (2021), pp. 218–224. DOI: 10.1016/j.annemergmed.2020.09.453.
- [19] L. Chen, A. Amaize, and D. Barath. "Barriers to sustaining telemedicine during COVID-19 in rural areas". In: *Journal of Primary Care and Community Health* 11 (2020), p. 2150132720957356. DOI: 10.1177/2150132720957356.
- [20] A. K. Ngugi and P. K. Mbugua. "Community health workers and telemedicine in rural Kenya: Overcoming infrastructure barriers". In: *BMC Health Services Research* 21.1 (2021), p. 456. DOI: 10.1186/s12913-021-06478-9.
- [21] R. Chandrasekaran, B. Sankaranarayanan, and J. Pendergrass. "Unfulfilled promises of health information exchange: What inhibits ambulatory clinics from electronically sharing health information?" In: *International Journal of Medical Informatics* 149 (2021), p. 104428. DOI: 10.1016/j.ijmedinf.2021.104428.
- [22] E. P. Soriano and L. A. Martinez. "Perspectives on telemedicine across urban, rural and remote areas in the Philippines during the COVID-19 pandemic". In: *BMJ Open* 14.8 (2024), e084425. DOI: 10.1136/bmjhci-2023-100837.
- [23] A. C. Smith. "Data privacy and security concerns in electronic health records: Implications for patient trust". In: *Journal of Medical Internet Research* 23.6 (2021), e26789. DOI: 10.2196/26789.
- [24] A. V. G. Noceda et al. "Patient satisfaction with telemedicine in the Philippines during the COVID-19 pandemic: A mixed methods study". In: *BMC Health Services Research* 23.1 (2023), p. 277. DOI: 10.1186/s12913-023-09127-x.
- [25] M. T. Salazar. "Patient satisfaction with telemedicine consultations in the Philippines: A cross-sectional study". In: *Philippine Journal of Allied Health Sciences* 5.1 (2021), pp. 34–47.

- [26] M. L. Aquino, R. P. Santos, and J. A. Cruz. “Acceptance and utilization of telemedicine services among senior citizens in the Philippines during the COVID-19 pandemic”. In: *Philippine Journal of Health Research and Development* 26.2 (2022), pp. 45–58.
- [27] J. Adler-Milstein and E. Pfeifer. “Pre-pandemic assessment: A decade of progress in electronic health record adoption among U.S. hospitals”. In: *Health Affairs Scholar* 1.5 (2023), qxad056. DOI: 10.1093/haschl/qxad056.
- [28] C. A. Rhoades, B. E. Whitacre, and A. F. Davis. “Early adoption of telehealth/remote patient monitoring and hospital revenue changes during COVID-19”. In: *Journal of Telemedicine and Telecare* 31.10 (2025), pp. 1456–1468.
- [29] R. A. Tiu. “User-friendly design considerations for Filipino telemedicine platforms: A usability study”. In: *Asian Journal of Technology and Innovation* 30.3 (2022), pp. 412–429. DOI: 10.1080/19761597.2021.1987654.
- [30] S. O’Connor et al. “Barriers and facilitators to patient and public engagement and recruitment to digital health interventions: Protocol of a systematic review of qualitative studies”. In: *BMJ Open* 11.5 (2021), e048801. DOI: 10.1136/bmjopen-2021-048801.
- [31] E. Gill et al. “Technology-facilitated care coordination in rural areas: What is needed?”. In: *International Journal of Medical Informatics* 137 (2020), p. 104102. DOI: 10.1016/j.ijmedinf.2020.104102.
- [32] P. K. Betancor et al. “Efficient patient care in the digital age: Impact of online appointment scheduling in a medical practice and university hospital on the ‘no-show’ rate”. In: *Frontiers in Digital Health* 7 (2024), p. 1567397. DOI: 10.3389/fdgth.2024.1567397.
- [33] D. Kucuk, E. Ozkan, and A. Yilmaz. “A solution to reduce the impact of patients’ no-show behavior on hospital operating costs: Artificial intelligence-based appointment system”. In: *Healthcare* 12.21 (2024), p. 2161. DOI: 10.3390/healthcare12212161.
- [34] M. Mazaheri, P. Sadeghi, and M. Ahmadi. “Evaluation of no-show rate in outpatient clinics with open access scheduling system: A systematic review”. In: *Health Science Reports* 7.7 (2024), e2169. DOI: 10.1002/hsr2.2169.
- [35] D. B. Ferro et al. “Improving healthcare access management by predicting patient no-show behavior”. In: *Decision Support Systems* 138 (2020), p. 113398. DOI: 10.1016/j.dss.2020.113398.
- [36] L. F. Dantas et al. “No-shows in appointment scheduling: A systematic literature review”. In: *Health Policy* 122.4 (2018), pp. 412–421. DOI: 10.1016/j.healthpol.2018.02.002.
- [37] C. M. Cusack et al. *Practice-based population health: Information technology to support transformation to proactive primary care*. Tech. rep. Agency for Healthcare Research and Quality, 2020.

- [38] P. J. Acacio-Claro, M. R. J. E. Estuar, and D. A. R. Villamor. “Understanding adoption of electronic medical records during a health emergency: An analysis of EMR usage logs from rural health facilities in the Philippines”. In: *Acta Medica Philippina* 58.22 (2024), pp. 7–13.
- [39] F. A. Mendoza, M. L. Cruz, and R. P. Santos. “Electronic medical records adoption in resource-constrained healthcare facilities: Challenges and success factors in the Philippines”. In: *Health Information Science and Systems* 11.1 (2023), p. 18. DOI: 10.1007/s13755-023-00218-4.
- [40] K. C. Derecho et al. “Technology adoption of electronic medical records in developing economies: A systematic review on physicians’ perspective”. In: *DIGITAL HEALTH* 10 (2024), p. 20552076231224605. DOI: 10.1177/20552076231224605.
- [41] L. M. Reyes, A. B. Santos, and J. P. Cruz. “Healthcare provider information systems and referral coordination in Philippine rural health units”. In: *International Journal of Health Planning and Management* 38.2 (2023), pp. 456–471. DOI: 10.1002/hpm.3589.
- [42] R. G. Hernandez and S. Becker. “Healthcare provider databases and their impact on clinical efficiency in low-resource settings”. In: *Health Information Management Journal* 51.2 (2022), pp. 89–102. DOI: 10.1177/18333583211012345.
- [43] S. Y. Patel et al. “Trends in outpatient care delivery and telemedicine during the COVID-19 pandemic in the US”. In: *JAMA Internal Medicine* 183.1 (2023), pp. 44–54. DOI: 10.1001/jamainternmed.2022.5644.
- [44] J. Wang et al. “Digital appointment systems and patient flow optimization in underserved healthcare facilities”. In: *International Journal for Quality in Health Care* 35.2 (2023), mzad023. DOI: 10.1093/intqhc/mzad023.
- [45] A. M. Del Rosario. “Digital appointment scheduling systems in Philippine rural health centers: Impact on waiting times and patient satisfaction”. In: *Philippine Journal of Public Health* 15.1 (2024), pp. 78–92.
- [46] M. Ramirez et al. “Automated remote monitoring of depression: Acceptance among low-income patients in diabetes disease management”. In: *JMIR Mental Health* 8.4 (2021), e24541. DOI: 10.2196/24541.
- [47] B. Holtz et al. “Using the Technology Acceptance Model to characterize barriers and opportunities of telemedicine in rural populations: Survey and interview study”. In: *JMIR Formative Research* 6.4 (Apr. 2022), e35130. DOI: 10.2196/35130.
- [48] A. K. S. Ong et al. “Factors influencing the acceptance of telemedicine in the Philippines”. In: *Technology in Society* 70 (2022), p. 102040. DOI: 10.1016/j.techsoc.2022.102040.
- [49] M. O. Gallardo, J. Dela Torre, and R. Ebardo. “The role of initial trust in the behavioral intention to use telemedicine among Filipino older adults”. In: *Gerontology and Geriatric Medicine* 10.1 (2024). DOI: 10.1177/2333721423122211.

- [50] M. M. Kepper et al. “Expanding a behavioral view on digital health access: Drivers and strategies to promote equity”. In: *Journal of Medical Internet Research* 26 (2024), e51355. DOI: 10.2196/51355.
- [51] X. Li et al. “Enabling telemedicine from the system-level perspective: Scoping review”. In: *Journal of Medical Internet Research* (2025).
- [52] International Organization for Standardization. *ISO 9241-210:2019 Ergonomics of human-system interaction—Part 210: Human-centred design for interactive systems*. <https://www.iso.org/standard/77520.html>. 2019.
- [53] Centers for Disease Control and Prevention. *About rural health*. <https://www.cdc.gov/rural-health/php/about/index.html>. 2025.
- [54] S-Pro. *Components of telemedicine: System architecture overview for 2024*. <https://s-pro.io/blog/telemedicine-system-components-architecture>. 2024.
- [55] eBusiness@Newcastle. *Task-Technology Fit — TheoryHub — Academic theories reviews for research and T&L*. <https://open.ncl.ac.uk/theories/3/task-technology-fit/>.
- [56] D. Marikyan and S. Papagiannidis. *Technology Acceptance Model: A review*. <http://eprints.ncl.ac.uk/302879>. Newcastle University. 2024.
- [57] National Economic and Development Authority Regional Office 5. *Bicol Regional Development Plan 2023–2028*. <https://rdc5.gov.ph/bicol-regional-development-plan-rdp-2023-2028/>. 2023.
- [58] H. K. Almathami, K. T. Win, and E. Vlahu-Gjorgievska. “Barriers and facilitators that influence telemedicine-based, real-time, online consultation at patients’ homes: Systematic literature review”. In: *Journal of Medical Internet Research* 22.2 (2020), e16407. DOI: 10.2196/16407.
- [59] L. W. Chang et al. “Impact of a mHealth intervention for peer health workers on AIDS care in rural Uganda: A mixed methods evaluation of a cluster-randomized trial”. In: *AIDS and Behavior* 26.1 (2022), pp. 25–36. DOI: 10.1007/s10461-021-03316-0.
- [60] A. Garavand et al. “Acceptance of health information technologies, acceptance of mobile health: A review article”. In: *Frontiers in Health Informatics* 12.1 (2023), p. 128. DOI: 10.30699/fhi.v12i1.456.
- [61] B. F. Kocyigit and T. Bal. “Factors affecting the acceptance of mobile health applications: A systematic review”. In: *Health Informatics Journal* 28.2 (2022), p. 14604582221099259. DOI: 10.1177/14604582221099259.
- [62] J. Y. Lee et al. “Using telemedicine to support care for people with type 2 diabetes mellitus: A qualitative analysis of patients’ perspectives”. In: *BMJ Open* 10.10 (2020), e042396. DOI: 10.1136/bmjopen-2020-042396.
- [63] MGMA Research Team. *Shortening wait times for appointments: Patient access strategies for 2025*. MGMA Stat. Dec. 2024.

- [64] S. W. Ong et al. “Telemedicine in low- and middle-income countries: A scoping review”. In: *Cureus* 14.7 (2022), e26808. DOI: 10.7759/cureus.26808.
- [65] F. Wurster et al. “Roger’s diffusion of innovations theory and the adoption of a patient portal’s digital anamnesis collection tool: Study protocol for the MAiBest project”. In: *Implementation Science Communications* 5.1 (2024), p. 74. DOI: 10.1186/s43058-024-00614-8.
- [66] J. Lopez. “Impact of online appointment scheduling on patient satisfaction in ophthalmology practice”. In: *International Journal of Healthcare Management* 18.1 (2025), pp. 45–58. DOI: 10.1080/20479700.2024.2345678.
- [67] A. Thanthrige et al. “Determinants of health care technology adoption using an integrated unified theory of acceptance and use of technology and task technology fit model: Systematic review and meta-analysis”. In: *Journal of Medical Internet Research* (2025). DOI: 10.2196/64524.
- [68] D. Tsogbe et al. “Adopting telehealth for mental health support in stroke rehabilitation in Sub-Saharan Africa: A conceptual analysis using Rogers’ Diffusion of Innovation Theory”. In: *PLOS Global Public Health* 6.2 (Feb. 2026), e0005025. DOI: 10.1371/journal.pgph.0005025.
- [69] A. Haleem et al. “Telemedicine for healthcare: Capabilities, features, barriers, and applications”. In: *Smart Health* (2021), p. 100117. DOI: 10.1016/j.sintl.2021.100117.

Curriculum Vitae

Appendices